

HW5 - Kernel Methods

Vanja Stojanović

May 2026

1 Kernels on the sine data set

200 noisy observations over $x \in [0, 19.9]$: three periods of a sine of wavelength ≈ 6.3 shifted to $\bar{y} \approx 1.5$. The polynomial Gram $(1+xx')^M$ grows like x^{2M} and is ill-conditioned on raw x , so we standardize x to zero mean unit variance before fitting (kernel formula untouched). **Selection procedure:** per parameter, swept a short grid, picked the value at which the fit visually transitioned from underfit to overfit, and sanity-checked training MSE. The noise scale used below is $\hat{\sigma}_{\text{noise}} \approx 0.3$, estimated from the residual std of the KRR + RBF fit. **Support vector definition:** a training point is an SV iff $|\alpha_i - \alpha_i^*| > 10^{-5}$; by KKT, points strictly inside the ε -tube have $\alpha_i = \alpha_i^* = 0$, and the 10^{-5} is a numerical tolerance against QP solver noise.

Table 1: Per-parameter sweep. Picked value in **bold**.

parameter	value	observation
RBF σ	1.0	underfits, averages across half-periods
	0.5	clean fit, indistinguishable from 0.25; pick larger for stronger smoothing
	0.25	visually identical to 0.5
	0.1	tracks individual noisy samples
Poly M	6	misses the third oscillation
	8	flattens the last peak
	11	tracks all three oscillations
	15	worsens edge blowup, no interior gain
KRR λ	10^{-6}	indistinguishable from 10^{-3}
	10^{-3}	default well-conditioned regime
	10^{-1}	flattens the fit
SVR ε	0.2	≈ 70 SVs, no real sparsity
	0.5	$\approx 1.5\hat{\sigma}_{\text{noise}}$, ~ 10 SVs, clean fit
	1.0	flattens oscillation amplitude
SVR λ	10^{-3}	$C=1000$, poly SVR overshoots interior peaks
	10^{-2}	$C=100$, balanced
	10^{-1}	$C=10$, flattens

Table 2: Final fit summary at picked hyperparameters. $\text{std}(y) \approx 1.07$.

method	kernel	λ	ε	MSE	#SV
KRR	RBF	10^{-3}		0.076	
KRR	Polynomial	10^{-3}		0.079	
SVR	RBF	10^{-2}	0.5	0.079	11 / 200
SVR	Polynomial	10^{-2}	0.5	0.085	16 / 200

Findings. All four MSEs sit in $[0.076, 0.085]$, residual std $\approx 0.28 \approx \hat{\sigma}_{\text{noise}}$: at these settings every configuration is noise-saturated. SVR achieves the same fit quality as KRR with $\approx 18\times$ (RBF) and $12.5\times$ (poly) fewer prediction-time terms; the circled SVs sit at or just outside the tube boundary, the geometric signature of complementary slackness. Both polynomial fits diverge at $x \rightarrow 20$: the Runge mechanism, a degree-11 monomial basis amplifies any boundary discrepancy as x^{11} ; RBF avoids this because every basis function decays away from its centre, so beyond the training support the prediction relaxes to the constant b . λ shrinks α in both methods, but in SVR it additionally caps $|\alpha_i^{(*)}| \leq C = 1/\lambda$: small λ lets one nearby SV dominate a peak, large λ spreads the load and smooths.

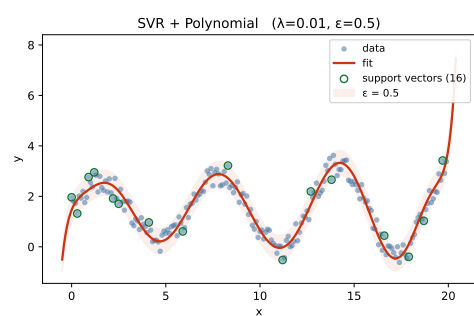
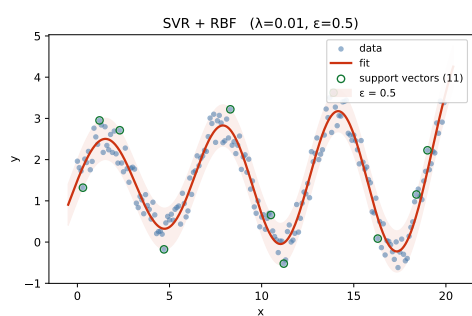
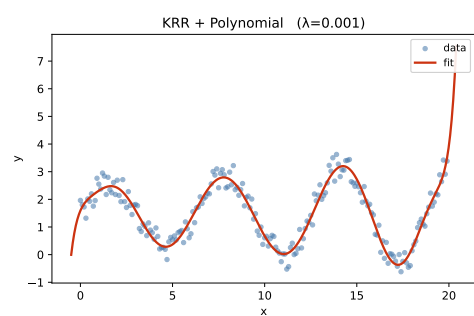
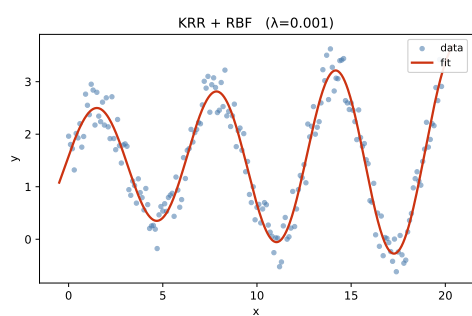


Figure 1: Fits on the sine data. Top: KRR + RBF / Polynomial. Bottom: SVR + RBF / Polynomial. Support vectors are circled; the shaded band is the $\pm\epsilon$ tube around the SVR fit.

2 Kernels on housing2r

200 observations, 5 features (Boston-housing subset), target y (median home value) with $\bar{y} \approx 21.7$, $\text{std}(y) \approx 7.9$. Single 80/20 train/test split, fixed seed. Features are standardized using train statistics only (std ranges from 0.7 to 166, raw values would dominate the Gram matrix). For each kernel parameter we report two test-MSE curves: $\lambda = 1$ fixed, and λ chosen by 5-fold CV on the train set over $\lambda \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$. SVR uses a single $\varepsilon = 4.6$, set from the residual std of a probe KRR + RBF fit ($\hat{\sigma}_{\text{noise}} \approx 4.55$); this is $\approx 0.6 \text{std}(y)$, narrow enough to track signal and wide enough to leave most points inside the tube.

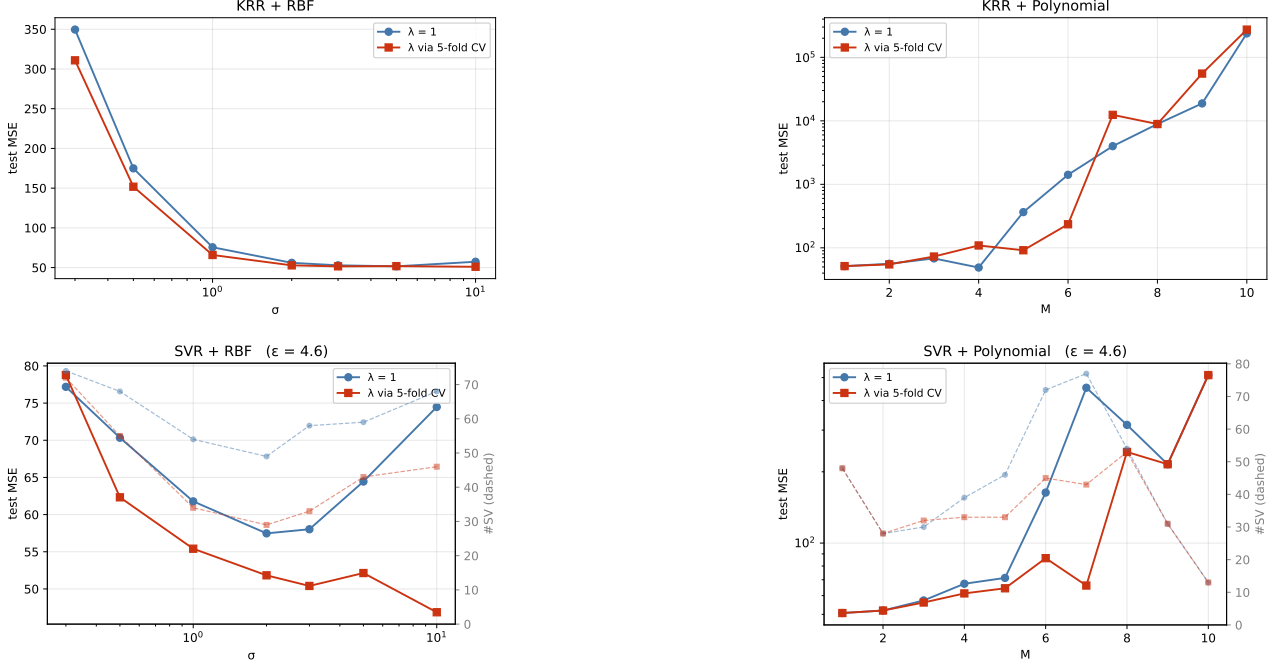


Figure 2: Test MSE vs. kernel parameter on housing2r. Top: KRR + RBF (left, linear y) and KRR + Polynomial (right, log y). Bottom: SVR with both kernels; dashed lines show #SV on the right axis.

Table 3: Best test MSE per (method, kernel). λ^* is the inner-CV choice at the best kernel parameter.

method	kernel	best param	λ^*	best MSE	#SV
KRR	RBF	$\sigma = 10$	10^{-3}	51.0	
KRR	Polynomial	$M = 1$	10^{-2}	51.5	
SVR	RBF	$\sigma = 10$	10^{-3}	46.9	46 / 160
SVR	Polynomial	$M = 1$	1.0	50.6	48 / 160

Polynomial kernel. KRR + Poly blows up past $M \approx 6$ (Fig. 2 top-right, log-y): on standardized inputs the Gram entries scale like $\|x\|^{2M} \sim 5^M$ in $d = 5$, the matrix becomes catastrophically ill-conditioned, and $(K + \lambda I)^{-1}$ amplifies any small-eigenvalue direction. CV- λ helps only at $M \leq 6$. SVR + Poly tolerates the same M much better because the box constraint $|\alpha_i^{(*)}| \leq C$ caps the influence of any single direction. Both polynomial families peak at $M = 1$ (essentially linear ridge with a bias).

RBF kernel. Smooth U-shape for both methods with minimum at $\sigma \in [3, 10]$. With standardized 5-D inputs, typical pairwise distances are $\sqrt{2d} \approx 3$, so σ in the range $[3, 10]$ is the regime where the kernel sees the data globally rather than treating each point as its own cluster. Small σ overfits, large σ underfits.

Effect of CV- λ . The CV curve never loses to $\lambda = 1$ and is materially better at the extremes of every sweep, where the fixed $\lambda = 1$ is either too strong (smooths out signal) or too weak (lets ill-conditioning bite). At the best $\sigma = 10$, SVR + RBF goes from MSE 74.5 to 46.9, a 37% reduction. Inner CV adds negligible compute on this data set.

KRR vs. SVR. Comparison axes used in the literature: (i) predictive accuracy, (ii) outlier robustness, (iii) predict-time sparsity, (iv) hyperparameter / solver complexity. Here SVR + RBF beats KRR + RBF by $\sim 8\%$ MSE at their respective optima. SVR’s ε -insensitive loss is piecewise-linear outside the tube rather than quadratic, so a single far outlier contributes $|y - \hat{y}|$ rather than $(y - \hat{y})^2$ to the gradient (documented robustness advantage; relevant on housing where a handful of homes have $y = 50$). SVR predicts in $O(\text{\#SV}) = 46$ vs.

$O(n) = 160$ for KRR. KRR has one hyperparameter and a closed-form fit; SVR has two and requires a QP solver.

Preference: SVR + RBF with inner-CV λ . It gives the best test MSE (46.9 vs. KRR 51.0), a $3.5\times$ sparser predictor (46 / 160), and a more robust loss against the heavy-tailed residuals in housing data. KRR is the right pick when one wants a single closed-form fit without a QP solver, accepting $\sim 8\%$ worse MSE in exchange for simpler workflow. The polynomial kernel is dominated by RBF on this data regardless of method.

3 A non-trivial kernel for structured data

Setup. Strings are structured data: order and gaps matter. We compare KRR with the *String Subsequence Kernel* (SSK) of Lodhi et al. 2002 against KRR on the strongest finite attribute tables one would naturally write down for strings. All methods share the same regressor and the same 5-fold inner CV for λ ; only the feature representation differs. Data: 300 strings of length 25 over $\Sigma = \{A, C, G, T\}$, target

$$y(s) = 1.0 \cdot N_{\text{ACT}}(s) - 0.5 \cdot N_{\text{GTA}}(s) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 0.3^2),$$

where $N_u(s)$ counts ordered occurrences of pattern u with span $\in [3, 8]$ (span 3 is contiguous, spans 4 to 8 are gapped). 80/20 train/test split, fixed seed.

Definition and why this is non-trivial. For length p and decay $\lambda_{\text{decay}} \in (0, 1]$,

$$\phi_u(s) = \sum_{\mathbf{i}: u=s[\mathbf{i}]} \lambda_{\text{decay}}^{l(\mathbf{i})}, \quad K_p(s, t) = \sum_{u \in \Sigma^p} \phi_u(s) \phi_u(t),$$

with $\mathbf{i} = (i_1 < \dots < i_p)$ and $l(\mathbf{i}) = i_p - i_1 + 1$ the index-tuple span. Computed by an $O(p|s||t|)$ DP without enumerating Σ^p .

For our $|\Sigma| = 4$ and $p = 3$, the feature space has $|\Sigma|^p = 64$ coordinates; for $p = 5$ it grows to 1024. SSK can be written as a finite table at small p , but each coordinate $\phi_u(s)$ is a gap-weighted DP output, not a count anyone would write down by inspection. The natural attribute representations for strings (k -mer counts) are strictly different from ϕ_u and cannot reproduce SSK regardless of regressor or regularization, since they encode no gap information.

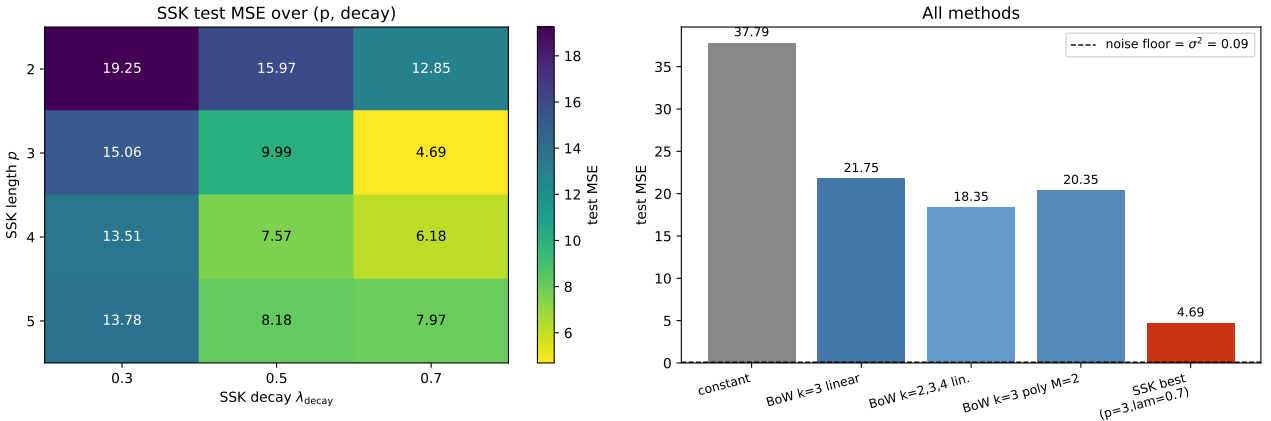


Figure 3: Left: SSK test MSE across the $(p, \lambda_{\text{decay}})$ grid; λ is chosen by 5-fold inner CV at each cell. Right: all methods compared. Dashed line is the irreducible noise floor $\sigma^2 = 0.09$.

Table 4: Test MSE on the synthetic string regression task. λ^* chosen by 5-fold inner CV; SSK rows show the four-by-three sweep summary.

class	representation	feature dim	λ^*	test MSE
trivial	constant (predict $\bar{y}_{\text{train}}$)			37.79
table	bag-of-3-mers + linear kernel	64	10^1	21.75
table	bag-of- k -mers ($k = 2, 3, 4$) concat + linear	336	10^1	18.35
table	bag-of-3-mers + Polynomial($M = 2$)	64	10^1	20.35
SSK	raw strings, worst cell in sweep ($p = 2, \lambda_d = 0.3$)		10^{-1}	19.25
SSK	raw strings, median over the 4×3 sweep		varies	10.99
SSK	raw strings, best cell ($p = 3, \lambda_d = 0.7$)		10^{-2}	4.69
irreducible noise floor $\sigma^2 = 0.3^2$				0.09

Robustness sweep ($p \in \{2, 3, 4, 5\}$, $\lambda_{\text{decay}} \in \{0.3, 0.5, 0.7\}$). The heatmap in Fig. 3 shows the result is robust across hyperparameters: **11 of 12 cells** beat the strongest table baseline (18.35), and only the single worst cell ($p = 2, \lambda_d = 0.3$, MSE 19.25) is comparable to it. MSE improves monotonically with λ_{decay} at every p (larger decay gives slower weight falloff and so better captures span 4 to 8 occurrences), and the optimum is at $p = 3$ (matching the pattern length used in the target) with $\lambda_d = 0.7$ (largest tested decay).

Strengthened baselines. The natural attribute representations were stacked in increasing flexibility: bag-of-3-mers linear (21.75), then bag-of- k -mers for $k \in \{2, 3, 4\}$ concatenated (18.35, $5.25\times$ more features), then bag-of-3-mers with a polynomial kernel of degree 2 (20.35, adding implicit pairwise count interactions). None goes below ≈ 18 , while SSK at its best cell achieves 4.69. Table-based predictors can only capture the contiguous span-3 component of the target signal; the span 4 to 8 component is out of reach for any function of k -mer counts.

Conclusion. Explicit string-structure modelling reduces MSE by $\approx 4\times$ versus the best feasible table (4.69 vs 18.35). The same KRR machinery from Parts 1 and 2 is reused; only the kernel object changes.